

# Cheat Sheet: Python Data Structures Part-2

## Dictionaries

| Package/Method        | Description                                                                                                                                                                                                                                                                                                                                            | Code Example                                                                                                                                                                                     |
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| Creating a Dictionary | A dictionary in Python is a built-in data structure that stores information in key-value pairs. Each key must be unique, and it is used to access its corresponding value. Dictionaries are enclosed in curly braces {}, and each pair is separated by a comma.                                                                                        | Example:<br><br>dict_name = {} #Creates an empty dictionary<br>person = { "name": "John", "age": 30, "city": "New York"}                                                                         |
| Accessing Values      | You can access the values in a dictionary using their corresponding keys.                                                                                                                                                                                                                                                                              | Syntax:<br><br>Value = dict_name["key_name"]<br><br>Example:<br><br>name = person["name"]<br>age = person["age"]                                                                                 |
| Add or modify         | You can add or modify elements in a dictionary by assigning a value to a key using the assignment operator =. When you assign a value to a key that does not already exist, Python automatically creates a new key-value pair and adds it to the dictionary. However, if the key already exists, the existing value will be replaced with the new one. | Syntax:<br><br>dict_name[key] = value<br><br>Example:<br><br>person["Country"] = "USA" # A new entry will be created.<br>person["city"] = "Chicago" # Update the existing value for the same key |
| del                   | Removes the specified key-value pair from the dictionary. Raises a KeyError if the key does not exist.                                                                                                                                                                                                                                                 | Syntax:<br><br>del dict_name[key]                                                                                                                                                                |

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|               |                                                                                                                                   | <p>Example:</p> <pre>del person["Country"]</pre>                                                                                  |
| key existence | You can check for the existence of a key in a dictionary using the <code>in</code> keyword                                        | <p>Example:</p> <pre>if "name" in person:<br/>    print("Name exists in the dictionary.")</pre>                                   |
| keys()        | Retrieves all keys from the dictionary and converts them into a list. Useful for iterating or processing keys using list methods. | <p>Syntax:</p> <pre>keys_list = list(dict_name.keys())</pre> <p>Example:</p> <pre>person_keys = list(person.keys())</pre>         |
| values()      | Extracts all values from the dictionary and converts them into a list. This list can be used for further processing or analysis.  | <p>Syntax:</p> <pre>values_list = list(dict_name.values())</pre> <p>Example:</p> <pre>person_values = list(person.values())</pre> |

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| items() | Retrieves all key-value pairs as tuples and converts them into a list of tuples. Each tuple consists of a key and its corresponding value. | <p>Syntax:</p> <pre>items_list = list(dict_name.items())</pre> <p>Example:</p> <pre>info = list(person.items())</pre> |

Sets

| Package/Method | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Code Example                                                                                                                                                                                                                |
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| add()          | Add elements to a set. You can use the add() method to insert a new element into a set. If the element already exists, it will not be added again, since sets automatically store only unique values.                                                                                                                                                                                                                                                                          | <p>Syntax:</p> <pre>set_name.add(element)</pre> <p>Example:</p> <pre>fruits.add("mango")</pre>                                                                                                                              |
| Defining Sets  | <p>A set in Python is an unordered collection that stores unique elements, meaning it automatically removes duplicates. Sets are enclosed in curly braces {} and are commonly used when you need to store a group of distinct values or perform mathematical operations such as union, intersection, and difference.</p> <p>Because sets are unordered, the elements have no fixed position or index, and their order may change each time you print or iterate over them.</p> | <p>Example:</p> <pre>empty_set = set() #Creating an Empty Set fruits = {"apple", "banana", "orange"} colors = ("orange", "red", "green")</pre> <p><b>Note:</b> These two sets will be used in the examples that follow.</p> |
| in             | Checks if an element exists in a set. Use the in keyword to verify whether a specific element is present in a set. It returns True if the element exists, and False otherwise.                                                                                                                                                                                                                                                                                                 | <p>Syntax:</p> <pre>element in set_name</pre>                                                                                                                                                                               |

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|              |                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Example:</p> <pre>"banana" in fruits</pre>                                                                                    |
| issubset()   | <p>Check if one set is a subset of another. The <code>issubset</code> method is used to determine whether all elements of one set exist within another set. If every element of the current set is found in the specified set, it returns <code>True</code>; otherwise, it returns <code>False</code>.</p>                                                                                                            | <p>Syntax:</p> <pre>is_subset = set1.issubset(set2)</pre> <p>Example:</p> <pre>is_subset = fruits.issubset(colors)</pre>         |
| issuperset() | <p>Check if one set is a superset of another. The <code>issuperset()</code> method is used to verify whether the current set contains all elements of another set. It returns <code>True</code> if every element of the specified set exists within the current set; otherwise, it returns <code>False</code>.</p>                                                                                                    | <p>Syntax:</p> <pre>is_superset = set1.issuperset(set2)</pre> <p>Example:</p> <pre>is_superset = colors.issuperset(fruits)</pre> |
| remove()     | <p>Remove a specific element from a set. The <code>remove()</code> method is used to delete a particular element from a set. If the element you try to remove does not exist, Python will raise a <code>KeyError</code>, which means you must ensure the element is present before calling this method.</p> <p>This method is useful when you are certain that the item you want to remove is already in the set.</p> | <p>Syntax:</p> <pre>set_name.remove(element)</pre>                                                                               |

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|                |                                                                                            | <p>Example:</p> <pre>fruits.remove("banana")</pre>                                                                                                                                                                                                                                                                                                                                  |
| Set Operations | Perform various operations on sets: union, intersection, difference, symmetric difference. | <p>Syntax:</p> <pre>union_set = set1.union(set2) intersection_set = set1.intersection(set2) difference_set = set1.difference(set2) sym_diff_set = set1.symmetric_difference(set2)</pre> <p>Example:</p> <pre>combined = fruits.union(colors) common = fruits.intersection(colors) unique_to_fruits = fruits.difference(colors) sym_diff = fruits.symmetric_difference(colors)</pre> |



# Skills Network

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